

USER MANUAL: HOSI PHASEALIGN PRO

(v1.3)

Hosi PhaseAlign Pro is a JSFX utility plugin designed for REAPER. It is a professional tool built to solve phase cancellation and time-alignment issues frequently encountered when recording with multiple microphones (e.g., drum kits, guitar amps) or when combining DI and Amp signals (e.g., Bass).

1. KEY FEATURES

- **Absolute Phase Synchronization (Sub-sample Interpolation):** Utilizes the *Hermite Cubic* interpolation algorithm, allowing audio time-shifting at a sub-sample level without degrading high-frequency quality.
- **Auto-Learn (Smart Scanning):** A 2-step scanning algorithm (coarse and fine scan) automatically analyzes and finds the perfect delay to fully phase-align two signals.
- **Multiband Phase Processing:** Independent phase rotation for 3 frequency bands: Bass (<200Hz), Mid (200Hz - 2000Hz), and High (>2000Hz).
- **Bass Lock:** Forces the low-end (<200Hz) to Mono to ensure a solid bass foundation, preventing hollowness caused by L/R phase cancellation.
- **Professional Visual Analyzer:** Integrates a real-time Goniometer (XY Scope) and a Correlation Meter to help you visually monitor phase alignment.
- **Compact UI:** Automatically collapses the interface to show only the Analyzer screen when you shrink the plugin window vertically.

2. FEATURE GUIDE (MAIN INTERFACE)

A. GLOBAL ALIGN Section

This section affects the entire frequency spectrum of the audio.

1. **Samples Knob (-1000 to 1000):** Shifts the signal time in samples. Used for micro-adjustments.
2. **Time ms Knob (-50ms to 50ms):** Shifts the signal time in milliseconds. Used to compensate for delay caused by microphone distance (e.g., a Room mic is usually more delayed than a Close mic).
3. **Phase deg Knob (0° to 360°):** Overall phase rotation. Useful when audio is time-aligned but still exhibits slight phase shift due to mic/preamp characteristics.
4. **Polarity Button:** Inverts the audio signal polarity (equivalent to the "Ø" button on physical mixers). Flips the waveform by 180 degrees.
5. **BasLock (Bass Lock) Button:** When activated, keeps all audio below 200Hz completely in Mono.
6. **M.Check (Mono Check) Button:** Audition the mix in Mono format to check if any sounds cancel out or disappear.

7. **Monitor Mode Button (Stereo / Mono / L Only / R Only / Diff):** Switches monitoring modes. The *Diff (Difference - Side)* mode is highly useful for listening exclusively to the out-of-phase audio between the L and R channels.

B. MULTIBAND PHASE Section

Used to fix complex phase issues when audio is out-of-phase in a specific frequency band rather than globally.

Default Crossover: Bass/Mid point is 200Hz, Mid/High point is 2000Hz.

1. **Bass Knob:** Independent phase rotation for the low band (<200Hz).
2. **Mid Knob:** Independent phase rotation for the mid band (200Hz - 2000Hz).
3. **High Knob:** Independent phase rotation for the high band (>2000Hz). *Tip: Great for smoothing out harsh cymbal bleed spilling into other mics.*

C. ANALYZER & AUTO Section

1. **Goniometer (XY Scope Circular Display):** * Represents stereo audio visually. If the signal is a vertical straight line, the audio is Mono and in-phase.
 - If the signal is a horizontal line, the audio is completely out-of-phase (will cancel out if listened to in mono).
 - A signal spreading outward circularly indicates a wide, pleasant Stereo sound.
2. **Correlation Meter:** * Moves towards **Blue (+1)**: L and R signals are well in-phase (Good).
 - Stays in the **Middle (0)**: Wide stereo, little commonality between L and R.
 - Moves towards **Red (-1)**: Signals are canceling each other out. This needs immediate fixing by inverting Polarity or adjusting delay.
3. **AUTO LEARN Button:** The most valuable feature of the plugin. Automates the delay finding process.
4. **Preset Menu:** Ready-made settings (Snare, Kick, Vocal, Piano...) to quickly apply common mixing configurations.



3. UI INTERACTION TIPS

- **Reset Parameters:** Double-click any knob to return it to its original default value (0).
- **Fine-tune:** Hold Ctrl (on Windows) or Cmd (on Mac) and drag the mouse to turn the knob very slowly, allowing for extremely precise parameter adjustments.
- **Compact UI:** When you reduce the height of the plugin window in REAPER (drag the bottom window border up), all buttons will disappear, leaving only the Analyzer screen and a compact phase meter. Very handy to keep open on a secondary monitoring screen.



4. PRACTICAL APPLICATION EXAMPLES

Example 1: Aligning a Snare Drum (Top and Bottom Mics)

1. Insert PhaseAlign Pro on the Snare Bottom track.
2. Usually, the bottom and top heads of a snare drum move in opposite directions relative to

their respective microphone diaphragms.

3. Select the "**Snare**" Preset at the bottom right, or simply press the **Polarity** button.
4. You will immediately notice the Snare sound becomes thicker and has more "meat" (low-end).

Example 2: Using AUTO-LEARN for Bass (DI and Amp)

When recording Bass Guitar with a direct input (DI) and a mic in front of the speaker cabinet (Amp), the Amp signal is usually delayed because sound waves need time to travel through the air.

1. Send both the DI signal to the L channel and the Amp signal to the R channel of a common track (or route them accordingly in REAPER).
2. Insert PhaseAlign Pro on this grouped track.
3. Play the song at a section where the Bass is played most consistently.
4. Press the **AUTO LEARN** button. The button will change to SCANNING...
5. Wait a few seconds. The plugin will automatically analyze and turn the Samples knob to the exact number so both audio sources align 100%. The Correlation meter will turn blue and reach close to the +1 mark.

Developed by: Hosi | GUI Version 1.3